

The Rigid Pendulum: Theory, Approximation, and Simulation

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Github Project Link: <https://github.com/StevenCodeStack/Pendulum-Simulator>

Introduction

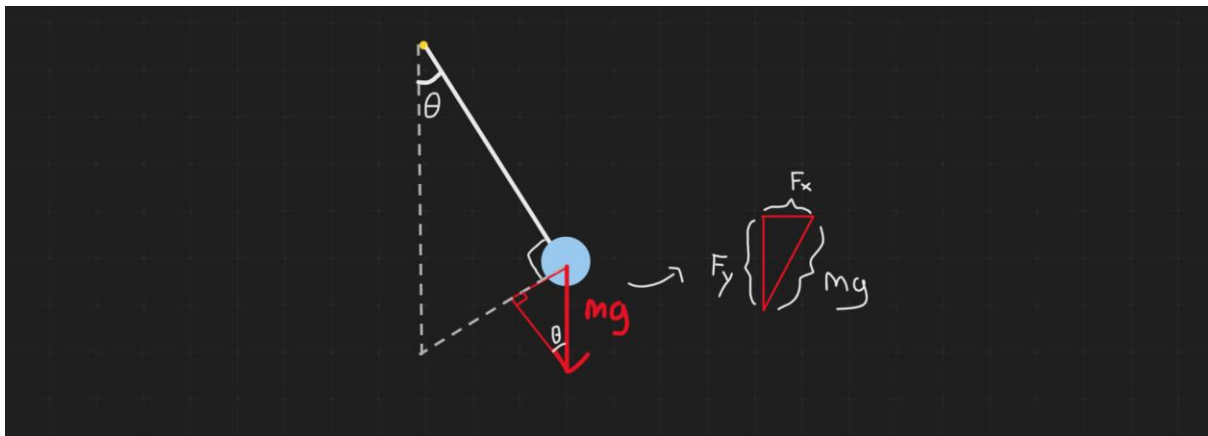
A project web-based rigid pendulum simulation where users can interact with it by adjusting variables, such as length, angle, and gravitational acceleration. This simulation also allows user to generate more than one pendulum, as long as their device is capable to run the program.

Why I built this project

To understand the mechanism of a pendulum, the difference between the approximate and exact equations and why it existed in the first place. I also want to understand how just with one equation can create a realistic pendulum simulation on a computer.

Theory & Derivation

Formula of angular acceleration



From this picture, shows a pendulum at an angle. The gravitational force pointing downwards, but in the frame of reference of the Bob, only F_x does the work to rotate the Bob. F_y does not do any work because it is cancelled by the tension of the rod. Because the angle of pendulum on y-axis is the same angle as the gravitational force shown in the diagram, the only work done by gravity is $mg\sin(\theta)$. By Newton's Second Law, $F = ma$, and the only force acting on the Bob is the gravitational force

$$ma = -mg \sin(\theta) \rightarrow a = -g \sin(\theta)$$

The minus sign is there because acceleration causes θ approaching 0

Next, the proof of $s = \theta \cdot r$ where s is the arclength, r is the radius, and θ is the angle of the arclength (has to be in radian). To prove this, we need to look at the formula of circle circumference.

$$C = 2\pi r$$

A full circle has angle 2π and circumference $2\pi r$. If you take a fraction x of the circle, the angle is $\theta = x \cdot 2\pi$, and the arc length is $s = x \cdot 2\pi r$. Substituting $x = \frac{\theta}{2\pi}$ into the arc length equation gives:

$$s = \frac{\theta}{2\pi} \cdot 2\pi r = \theta r$$

So, the arc length is just the angle multiplied by the radius. This only works if the angle is in radians, because radians are defined this way

Next, take the equation $s = \theta r$ and take the second derivative of it with respect to time.

$$\frac{d^2 s}{dt^2} = \frac{d^2}{dt^2} (\ell \cdot \theta)$$

ℓ is the length of the rod, equivalent to radius of a circle

The second derivative of s is acceleration, and ℓ is a constant, so the equation simplified is

$$a = \ell \cdot \alpha$$

α is the angular acceleration

Previous equation, stated that $a = -g \sin(\theta)$, substitute that

$$\ell \cdot \alpha = -g \sin(\theta) \rightarrow \alpha = -\frac{g}{L} \cdot \sin(\theta)$$

Formula of Kinetic Energy and Potential Energy on Pendulums

Kinetic Energy:

Take the derivative of the equation $s = \ell \cdot \theta$

$$v = \ell \cdot \omega$$

ω is angular speed

So, we substitute v into the kinetic energy formula

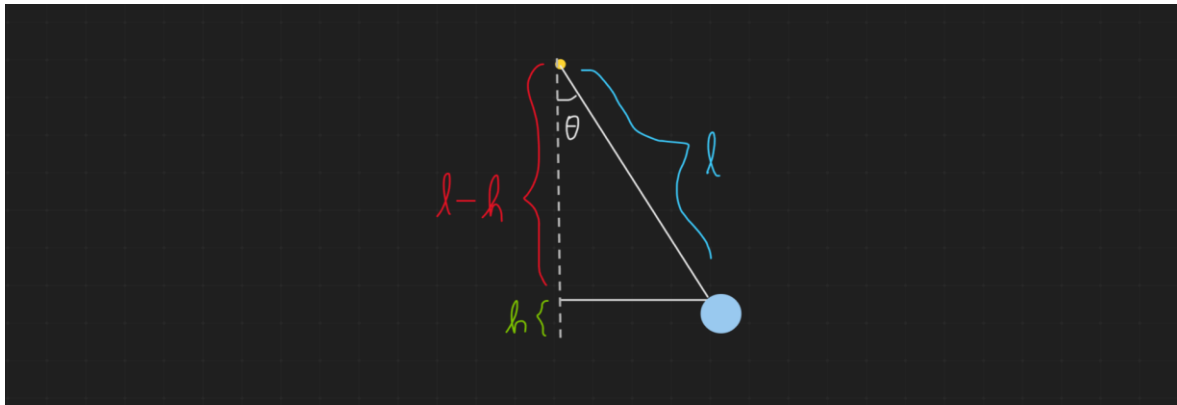
$$KE = \frac{1}{2} m v^2 = \frac{1}{2} m \ell^2 \omega^2$$

Potential Energy

The formula of potential energy is

$$PE = mgh$$

So, in pendulum scenario, we need to find the h



Looking at the picture, we see that

$$\cos(\theta) = \frac{l-h}{l}$$

Solve for h by isolating it

$$l \cdot \cos(\theta) = l - h \rightarrow h = l - l \cdot \cos(\theta) = l(1 - \cos(\theta))$$

Finally, we substitute h into the equation

$$PE = mg\ell(1 - \cos(\theta))$$

The Small-Angle Approximation: Derivation and Limitations

Previously, we have derived the equation of angular acceleration for all angle

$$\theta'' = -\frac{g}{L} \cdot \sin(\theta)$$

But in high school physics, it's simplified by assuming $\sin(\theta) = \theta$

$$\theta''(t) = -\frac{g}{L} \cdot \theta(t)$$

On small angles, where $0 < \theta < 15^\circ$, $\sin(\theta) \approx \theta$. Be able to plug in the angle instead of computing $\sin(\theta)$, where students usually only remember a specific angle pattern ($30^\circ, 45^\circ, 60^\circ, \dots$), is another small reason why it exists. But the main reason is to be able to solve $\theta(t)$.

The equation of motion for a pendulum is:

$$\theta''(t) = -\frac{g}{L} \sin(\theta(t))$$

This looks simple, but there the problem is the angle is inside the sine function.

To solve for $\theta(t)$, we would need to "undo" the sine function and find a formula that gives the angle at any time t . But because θ is inside $\sin$, we cannot isolate it using basic algebra or calculus. Try to integrate the equation twice

$$\theta(t) = \int \int -\frac{g}{L} \cdot \sin(\theta(t)) dt dt$$

The function $\theta(t)$ is unknown, so we cannot integrate it. We would need to use "Elliptical Integral" in order to find the function.

However, the simplified version where $\sin(\theta) = \theta$, It is possible because now the function is linear. First, we have to ask ourselves, what function $\theta(t)$ has a second derivative that is proportional to itself? This means what function, if I take the second derivative of it, the function contains the original function. There are only 2 functions that do this, which are sine and cosine function.

$$\theta(t) = A \cos(\omega t) + B \sin(\omega t)$$

Note: ω here is not angular speed. It's natural frequency where $\omega = \sqrt{\frac{g}{L}}$. It's a constant and determines the speed of the pendulum.

So how did we get this equation in the first place? We guessed it, but not in a brute force or mindless way because we know the second derivative of $\sin(\theta)$ and $\cos(\theta)$ are $-\sin(\theta)$ and $-\cos(\theta)$ respectively. Let's try to prove if the equation really works by taking the second derivative of it

$$\frac{d}{dx} \theta(t) = -\omega A \sin(\omega t) + \omega B \cos(\omega t)$$

$$\frac{d}{dx} \theta'(t) = -\omega^2 A \cos(\omega t) - \omega^2 B \sin(\omega t)$$

$$\theta''(t) = -\omega^2 (A \cos(\omega t) + B \sin(\omega t))$$

We know that $\theta(t) = A \cos(\omega t) + B \sin(\omega t)$, so we can substitute that

$$\theta''(t) = -\omega^2 \cdot \theta(t)$$

The equation $\theta(t) = A \cos(\omega t) + B \sin(\omega t)$, the components have their own role for better intuitive understanding.

When $t = 0$, $\theta(0) = A \cdot 1 + B \cdot 0 = A$. So, this tells us that A is a starting angle (θ_0)

And if we switch to the first derivative equation, $\theta'(0) = -\omega A \cdot 0 + \omega B \cdot 1 = \omega B$. This tells us that $v_0 = \omega B \rightarrow B = \frac{v_0}{\omega}$ where $v_0 = \theta'(0)$. So, our final equation is:

$$\theta(t) = \theta_0 \cos(\omega t) + \frac{v_0}{\omega} \sin(\omega t)$$

We derived the solution for the small-angle equation:

$$\theta(t) = \theta_0 \cos(\omega t)$$

Where $\omega = \sqrt{\frac{g}{L}}$

Next, we're going to derive the Period (T)

Period is defined as how long it takes to complete one full oscillation. From the equation $\theta(t) = \theta_0 \cos(\omega t)$ (Initial speed = 0), the cosine function repeats itself when its argument increases by 2π . So, the period T is the time it takes for ωt to go from 0 to 2π :

$$\omega T = 2\pi$$

Solving for T :

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

Noticed anything missing? Correct, the angle. So, consider the same length, same gravitational acceleration, but different angle. With this formula, the periods are all the same, but we know that's not true. That's the limitations. The formulas (T and $\theta(t)$) are all constructed from an equation that only works for small angle.

For the exact equation $\theta'' = -\frac{g}{L} \sin \theta$, the period is:

$$T = 4 \sqrt{\frac{L}{g}} \cdot K\left(\sin\left[\frac{\theta_0}{2}\right]\right)$$

where $K(k)$ is the complete elliptic integral that I myself cannot explain. This formula cannot be evaluated using basic functions, so numerical methods (or a calculator) are needed. For small angles, it simplifies to the high school formula $T = 2\pi\sqrt{L/g}$.

Consider $\ell = 1m, g = 9.8m/s^2, m = 1kg$

Angle	Approximate Period	Real Period
10°	2.006 s	2.006 s
30°	2.006 s	2.040 s
60°	2.006 s	2.150 s
90°	2.006 s	2.370 s

The difference will get bigger with larger l , g , or angle

Energy Conservation

In a frictionless pendulum, energy is conserved

$$ME = PE + KE = \text{constant}$$

Using the formulas derived earlier, the simulation tracks the value of PE, KE, and total energy overtime.

Consider $\ell = 1\text{m}$, $g = 9.8\text{m/s}^2$, $\theta = 60^\circ$, $m = 1\text{kg}$

Time (s)	KE (J)	PE (J)	ME (J)
0.00 s	0.00 J	7.35 J	7.35 J
1.59 s	2.42 J	4.93 J	7.35 J
2.40 s	1.87 J	5.48 J	7.35 J
3.72 s	1.96 J	5.39 J	7.35 J
5.17 s	0.43 J	6.92 J	7.35 J

However, my simulation is not 100% accurate. The total energy oscillates between the exact value and about 0.01% below it. It never exceeds the exact value. To make the error more visible, I scaled up the gravitational acceleration to 100 m/s^2 with an angle of 90° and $L = 100\text{ m}$.

Time (s)	KE (J)	PE (J)	ME (J)
0.00 s	0.00 J	10000.00 J	10000.00 J
1.17 s	6272.68 J	3727.28 J	9999.96 J
2.38 s	7618.21 J	2381.73 J	9999.94 J
3.46 s	329.54 J	9670.46 J	10000.00 J
5.50 s	9987.65 J	12.25 J	9999.90 J

Numerical Method

Why a computer is needed?

The angular acceleration on a pendulum is

$$\theta''(t) = -\frac{g}{L} \cdot \sin(\theta(t))$$

And we know we cannot find a clean $\theta(t)$ function. Although, the $\theta(t)$ function exists through elliptic integration, it's too complex for me. So, I decided to use the second method, which is numerical integration. With computer, this allows me to create a simulation that is easy and good enough.

The usual monitor refresh rate is 144Hz, I use that number to set $dt = 1/144$. If $dt = 1/1000$ or $1/10000$, it would give a more accurate result, however, the simulation is slowed down. The formula above gives the exact acceleration at θ . So after computing the acceleration, I multiplied that by dt to get the velocity and update the angular velocity. After

that, multiply the result by dt again to get displacement and update the new angle. The final formula is

$$\theta = \theta_0 + \omega_0 dt + a dt^2$$

But this uses Euler's method. It assumes the speed and acceleration are constant during the time step. In reality, the speed and acceleration change, so we should use the average speed and acceleration.

The exact position update uses the constant acceleration formula:

$$\theta_{n+1} = \theta_n + \omega_n \Delta t + \frac{1}{2} \alpha_n (\Delta t)^2$$

However, the velocity must also be updated correctly. In Euler's method, velocity is updated using only the initial acceleration, which adds energy to the system. To fix this, Verlet calculates the acceleration at the new position and uses the average of the old and new accelerations to update velocity:

$$\omega_{n+1} = \omega_n + \frac{1}{2} (\alpha_n + \alpha_{n+1}) \Delta t$$

The real difference is that Euler uses only the initial acceleration, while Verlet uses the average acceleration over the time step. Both Euler and Verlet move the pendulum from A to B in the same time step. However, Verlet uses the average of the old and new acceleration to update the velocity. This does not change the visual position during a single frame, but it prevents the slow energy drift that Euler's method causes, making the simulation more accurate over time

Conclusion

In this project, I derived the exact equation of motion for a pendulum, compared it with the high school small-angle approximation, and built a numerical simulation to test both equations. I also tracked the energy conservation to calculate how accurate the simulation is.

The result shows that small-angle approximation equation works well when angle is below 15° but fails at larger angles. Period depends on starting angle, which the small-angle formula does not account for.

The simulation uses numerical integration (Verlet's method) to solve the exact equation step by step. The energy conservation test shows that total mechanical energy stays constant to within 0.01%, confirming that the simulation is physically accurate.

This project helped me understand the difference between approximate and exact models in physics.